Pole Position

How does ADSL work? What is VDSL? When, and how, will true broadband reach your home? We take a brief look

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Who needs 5 Mbps? Of course you do, so you can *(illegally!)* download a movie in 20 minutes. But who needs 54 Mbps? You'd need to buy a new hard disk every day if you kept downloading at that speed! But streaming video, teleconferencing and even HDTV become possible at 54 Mbps, which is the theoretical speed offered by VDSL—Very-high-bit-rate DSL. VDSL is the hybrid (fibre-cum-twisted pair copper wire) evolution of DSL (Digital Subscriber Line). OK, it's sort of cheating to call it DSL, but then, much of the last mile is over telephone lines.

If you've lost us, refer *Turbocharging The Telephone*, *Digit* July 2004. There, we talked about DSL and how it works. Back then, DSL wasn't really available in India except for a few select cities; now, corporations such as BSNL, MTNL, and Airtel offer ADSL (Asynchronous DSL) in several cities. The services are very expensive for those of us who like to download a lot of stuff (and that's most of us!), but it's there. Plans from the companies mentioned above are pretty competetive for casual surfers.

In any case, our purpose here is to talk about what VDSL is and whether it's possible in India. But before that, a recap about how DSL—specifically ADSL—works.

ADSL

ADSL provides a bandwidth that differs from area to area, and between implementations; a typical figure would be 1.5 Mbps to 8 Mbps over a single copper pair, at distances up to 18,000 feet, and at a wire thickness of 0.5 mm.

You probably know that with ADSL, you can use your phone at the same time that you surf the Net. You're able to do this because the ADSL system divides the frequency spectrum into three parts—one for phone service, one for upstream data and one for downstream data.

Illustration Shrikrishna Patkar